

Doc Installation et configuration node.js

Installation de corosync et pacemaker

```
umah@node1:~$ sudo systemctl status pacemaker
● pacemaker.service - Pacemaker High Availability Cluster Manager
   Loaded: loaded (/lib/systemd/system/pacemaker.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-03-05 14:37:42 CET; 4 days ago
     Docs: man:pacemakerd
           https://clusterlabs.org/pacemaker/doc/
  Main PID: 492 (pacemakerd)
    Tasks: 7
   Memory: 64.1M
      CPU: 9min 20.193s
   CGroup: /system.slice/pacemaker.service
           └─492 /usr/sbin/pacemakerd
             └─494 /usr/lib/pacemaker/pacemaker-based
               └─495 /usr/lib/pacemaker/pacemaker-fenced
                 └─496 /usr/lib/pacemaker/pacemaker-execd
                   └─497 /usr/lib/pacemaker/pacemaker-attd
                     └─498 /usr/lib/pacemaker/pacemaker-schedulerd
                       └─499 /usr/lib/pacemaker/pacemaker-controld

mars 10 14:00:26 node1 pacemaker-controld[499]: notice: Transition 483 (Complete=0, Pending=0, Fired=0, Skipped=0, Incomplete=0, Source=/var/lib/pacemaker/pengine/pe-input-37.bz2)
mars 10 14:00:26 node1 pacemaker-controld[499]: notice: State transition S_TRANSITION_ENGINE -> S_IDLE
mars 10 14:15:26 node1 pacemaker-controld[499]: notice: State transition S_IDLE -> S_POLICY_ENGINE
mars 10 14:15:26 node1 pacemaker-schedulerd[498]: notice: Calculated transition 484, saving inputs in /var/lib/pacemaker/pengine/pe-input-37.bz2
mars 10 14:15:26 node1 pacemaker-controld[499]: notice: Transition 484 (Complete=0, Pending=0, Fired=0, Skipped=0, Incomplete=0, Source=/var/lib/pacemaker/pengine/pe-input-37.bz2)
mars 10 14:15:26 node1 pacemaker-controld[499]: notice: State transition S_TRANSITION_ENGINE -> S_IDLE
mars 10 14:30:26 node1 pacemaker-controld[499]: notice: State transition S_IDLE -> S_POLICY_ENGINE
mars 10 14:30:26 node1 pacemaker-schedulerd[498]: notice: Calculated transition 485, saving inputs in /var/lib/pacemaker/pengine/pe-input-37.bz2
mars 10 14:30:26 node1 pacemaker-controld[499]: notice: Transition 485 (Complete=0, Pending=0, Fired=0, Skipped=0, Incomplete=0, Source=/var/lib/pacemaker/pengine/pe-input-37.bz2)
mars 10 14:30:26 node1 pacemaker-controld[499]: notice: State transition S_TRANSITION_ENGINE -> S_IDLE
lines 1-28/28 (END)
```

```
boumah@node1:~$ sudo systemctl status corosync
● corosync.service - Corosync Cluster Engine
   Loaded: loaded (/lib/systemd/system/corosync.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-03-05 14:37:42 CET; 4 days ago
     Docs: man:corosync
           man:corosync.conf
           man:corosync_overview
  Main PID: 459 (corosync)
    Tasks: 9 (limit: 2306)
   Memory: 141.8M
      CPU: 35min 40.482s
   CGroup: /system.slice/corosync.service
           └─459 /usr/sbin/corosync -f

mars 10 14:35:47 node1 corosync[459]: [KNET ] rx: Packet rejected from 10.0.5.152:5405
mars 10 14:35:47 node1 corosync[459]: [KNET ] rx: Packet rejected from 10.0.5.152:5405
mars 10 14:35:48 node1 corosync[459]: [KNET ] rx: Packet rejected from 10.0.5.152:5405
mars 10 14:35:49 node1 corosync[459]: [KNET ] rx: Packet rejected from 10.0.5.152:5405
mars 10 14:35:50 node1 corosync[459]: [KNET ] rx: Packet rejected from 10.0.5.152:5405
mars 10 14:35:51 node1 corosync[459]: [KNET ] rx: Packet rejected from 10.0.5.152:5405
mars 10 14:35:51 node1 corosync[459]: [KNET ] rx: Packet rejected from 10.0.5.152:5405
mars 10 14:35:52 node1 corosync[459]: [KNET ] rx: Packet rejected from 10.0.5.152:5405
mars 10 14:35:53 node1 corosync[459]: [KNET ] rx: Packet rejected from 10.0.5.152:5405
mars 10 14:35:54 node1 corosync[459]: [KNET ] rx: Packet rejected from 10.0.5.152:5405
boumah@node1:~$
```

Ensuite je configure les nœuds et les active bien

IP première VM :

```
boumah@node2:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether bc:24:11:2d:88:0c brd ff:ff:ff:ff:ff:ff
    altname enp0s18
    inet 10.0.5.150/24 brd 10.0.5.255 scope global ens18
        valid_lft forever preferred_lft forever
    inet 10.0.5.155/24 brd 10.0.5.255 scope global secondary ens18:VIP
        valid_lft forever preferred_lft forever
    inet6 fe80::be24:11ff:fe2d:880c/64 scope link
        valid_lft forever preferred_lft forever
boumah@node2:~$ _
```

IP deuxième VM :

```
boumah@node1:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether bc:24:11:9e:90:8e brd ff:ff:ff:ff:ff:ff
    altname enp0s18
    inet 10.0.5.151/24 brd 10.0.5.255 scope global ens18
        valid_lft forever preferred_lft forever
    inet6 fe80::be24:11ff:fe9e:908e/64 scope link
        valid_lft forever preferred_lft forever
boumah@node1:~$ _
```

Je vérifie le status en faisant un crm status

```
boumah@node2:~$ sudo crm status
Status of pacemakerd: 'Pacemaker is running' (last updated 2026-03-11 11:04:32 +01:00)
Cluster Summary:
  * Stack: corosync
  * Current DC: node1 (version 2.1.5-a3f44794f94) - partition with quorum
  * Last updated: Wed Mar 11 11:04:32 2026
  * Last change: Wed Mar 11 11:04:07 2026 by root via cibadmin on node1
  * 2 nodes configured
  * 2 resource instances configured

Node List:
  * Online: [ node1 node2 ]

Full List of Resources:
  * Resource Group: servnode1:
    * IpVirtuelle (ocf:heartbeat:IPaddr2): Started node1
    * serviceWeb (lsb:apache2): Started node1
boumah@node2:~$
```

J'ai ensuite créer un moniteur pour la ressource avec des valeurs par défaut

```
boumah@boumah:/etc/corosync$ crm configure show
-bash: crm : commande introuvable
boumah@boumah:/etc/corosync$ sudo crm configure show
node 1: node1
node 2: node2
primitive IpVirtuelle IPAddr2 \
    params ip=10.0.5.155 cidr_netmask=255.255.255.0 nic=ens18 iflabel=VIP \
    op monitor interval=10s \
    op start timeout=30s interval=0 \
    op stop timeout=40s interval=0
property cib-bootstrap-options: \
    have-watchdog=false \
    dc-version=2.1.5-a3f44794f94 \
    cluster-infrastructure=corosync \
    cluster-name=debian \
    stonith-enabled=false
boumah@boumah:/etc/corosync$
```

```
boumah@boumah:/etc/corosync$ sudo systemctl restart corosync
boumah@boumah:/etc/corosync$ sudo systemctl restart pacemaker
boumah@boumah:/etc/corosync$ sudo crm status
Status of pacemakerd: 'Pacemaker is running' (last updated 2026-03-04 11:00:14 +01:00)
Cluster Summary:
  * Stack: corosync
  * Current DC: node1 (version 2.1.5-a3f44794f94) - partition with quorum
  * Last updated: Wed Mar  4 11:00:14 2026
  * Last change:  Wed Mar  4 10:59:44 2026 by root via cibadmin on node2
  * 2 nodes configured
  * 1 resource instance configured

Node List:
  * Online: [ node1 node2 ]

Full List of Resources:
  * IpVirtuelle (ocf:heartbeat:IPAddr2):      Started node1

boumah@boumah:/etc/corosync$ sudo ifconfig
ens18: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 10.0.5.150 netmask 255.255.255.0  broadcast 10.0.5.255
    inet6 fe80::be24:11ff:fe2d:880c prefixlen 64  scopeid 0x20<link>
    ether bc:24:11:2d:88:0c  txqueuelen 1000  (Ethernet)
    RX packets 168188  bytes 36976661 (35.2 MiB)
    RX errors 0  dropped 3822  overruns 0  frame 0
    TX packets 16927  bytes 1716509 (1.6 MiB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 1000  (Boucle locale)
    RX packets 0  bytes 0 (0.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 0  bytes 0 (0.0 B)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

boumah@boumah:/etc/corosync$
```

```

crm(live/node1)configure# group servnode1 IpVirtuelle serviceWeb meta migration-threshold="5"
INFO: modified location:cli-prefer-IPvirtuelle from IpVirtuelle to servnode1
crm(live/node1)configure# commit
WARNING: (unpack_config) warning: Blind faith: not fencing unseen nodes
crm(live/node1)configure# quit
bye
boumah@node1:~$ sudo crm status
Status of pacemakerd: 'Pacemaker is running' (last updated 2026-03-05 15:27:38 +01:00)
Cluster Summary:
* Stack: corosync
* Current DC: node1 (version 2.1.5-a3f44794f94) - partition with quorum
* Last updated: Thu Mar 5 15:27:38 2026
* Last change: Thu Mar 5 15:27:18 2026 by root via cibadmin on node1
* 2 nodes configured
* 2 resource instances configured

Node List:
* Online: [ node1 node2 ]

Full List of Resources:
* Resource Group: servnode1:
  * IpVirtuelle (ocf:heartbeat:IPaddr2): Started node1
  * serviceWeb (lsb:apache2): Started node1
boumah@node1:~$

```

Ensuite j'ai tester en essayant de mettre chacun des nœud en standby pour voir si ca marchait

```

boumah@node1:~$ sudo crm node standby node1
INFO: standby node node1
boumah@node1:~$ sudo crm status
Status of pacemakerd: 'Pacemaker is running' (last updated 2026-03-11 11:24:40 +01:00)
Cluster Summary:
* Stack: corosync
* Current DC: node1 (version 2.1.5-a3f44794f94) - partition with quorum
* Last updated: Wed Mar 11 11:24:40 2026
* Last change: Wed Mar 11 11:24:38 2026 by root via cibadmin on node1
* 2 nodes configured
* 2 resource instances configured

Node List:
* Node node1: standby
* Online: [ node2 ]

Full List of Resources:
* Resource Group: servnode1:
  * IpVirtuelle (ocf:heartbeat:IPaddr2): Started node2
  * serviceWeb (lsb:apache2): Started node2
boumah@node1:~$

```

```

boumah@node2:~$ sudo crm node standby node2
[sudo] Mot de passe de boumah :
INFO: "standby" is accepted as "standby"
INFO: standby node node2
boumah@node2:~$ sudo crm status
Status of pacemakerd: 'Pacemaker is running' (last updated 2026-03-11 11:27:04 +01:00)
Cluster Summary:
* Stack: corosync
* Current DC: node1 (version 2.1.5-a3f44794f94) - partition with quorum
* Last updated: Wed Mar 11 11:27:04 2026
* Last change: Wed Mar 11 11:26:59 2026 by root via cibadmin on node2
* 2 nodes configured
* 2 resource instances configured

Node List:
* Node node2: standby
* Online: [ node1 ]

Full List of Resources:
* Resource Group: servnode1:
  * IpVirtuelle (ocf:heartbeat:IPaddr2): Started node1
  * serviceWeb (lsb:apache2): Started node1
boumah@node2:~$

```